

Savannah L. Ferretti

✉ savannah.ferretti@uci.edu | 🌐 savannahferretti.github.io | 🔗 savannahferretti | 🐙 savannahferretti

EDUCATION

Ph.D. in Earth System Science, University of California, Irvine Jun 2026
M.S. in Earth System Science, University of California, Irvine Dec 2023
B.S. in Earth & Atmospheric Sciences, Cornell University Dec 2020

EXPERIENCE

Graduate Student Researcher Sep 2021–Jun 2026
Pritchard & Baldwin Labs, University of California, Irvine Irvine, CA

- Developed statistical & interpretable machine learning models to analyze variability in South Asian monsoon rainfall using large-scale reanalysis & satellite datasets
- Quantified uncertainty & evaluated predictive skill across model configurations for robustness
- Built open-source, scalable Python workflows for parallel processing in HPC environments
- Communicated findings through peer-reviewed publications & conference presentations

Consulting Analyst Mar–Aug 2021
Higher Education Research Services, Huron Consulting Group New York, NY

- Built Excel-based financial planning & reporting tools for a research nonprofit
- Developed long-term resource forecasts & presented recommendations to executive stakeholders

Undergraduate Research Assistant Jun–Aug 2020
Commene Lab, Lamont-Doherty Earth Observatory Palisades, NY

- Quantified changes in atmospheric methane concentrations during the NYC COVID-19 shutdown
- Applied meteorological analysis & transport modeling to attribute emissions to likely source regions

Student Administrative Assistant Jan–Dec 2020
CALS Office of Admissions, Cornell University Ithaca, NY

- Increased prospective student yield through creation of an interactive [ArcGIS campus tour](#)
- Improved student engagement through execution of an office-wide social media strategy

Undergraduate Research Assistant Sep 2019–May 2020
Pritchard Lab, Cornell University Ithaca, NY

- Processed satellite imagery to construct time series of volcanic thermal anomalies
- Investigated pre-eruptive thermal signals for inclusion in NASA's AVTOD volcano database

SKILLS

Python (PyTorch, PySR, Dask, Xarray, Pandas, Cartopy, Matplotlib) • Bash • Git/GitHub • High-Performance/Distributed Computing • NetCDF/HDF5/Zarr • Linux • Microsoft Excel (Certified Specialist) • $\text{\LaTeX}$ • WordPress

PUBLICATIONS

- S. L. Ferretti**, J. Lin, S. Shamekh, J. W. Baldwin, M. S. Pritchard, & T. Beucier. (2026). **Data-driven integration kernels for interpretable nonlocal operator learning**. In press at *Environmental Data Science*.
- S. L. Ferretti**, M. S. Pritchard, F. Ahmed, L. Peng, & J. W. Baldwin. (2025). **Explaining South Asian monsoon rainfall seasonality using a metric of plume buoyancy**. *Geophysical Research Letters*, 52(16), e2025GL115546.
- L. Peng, P. N. Blossey, W. M. Hannah, ...**S. L. Ferretti**, ...& M. S. Pritchard. (2025). **Resolving low cloud feedbacks globally with E3SM High-Res MMF: Agreement with LES but stronger shortwave effects**. *Journal of Advances in Modeling Earth Systems*, 17(6), e2025MS005003.
- S. Yu, Z. Hu, A. Subramaniam, ...**S. L. Ferretti**, ...& M. S. Pritchard. (2025). **ClimSim-Online: A large multi-scale dataset and framework for hybrid ML-physics climate emulation**. *Journal of Machine Learning Research*. 26(142), 1-85.
- S. Yu, W. M. Hannah, L. Peng, ...**S. L. Ferretti**, ...& M. S. Pritchard. (2023). **ClimSim: A large multi-scale dataset for hybrid physics-ML climate emulation**. *NeurIPS 2023 Datasets & Benchmarks Track*, **Outstanding Paper Award**.
- A. M. Jenney, **S. L. Ferretti**, & M. S. Pritchard. (2023). **Vertical resolution impacts explicit simulation of deep convection**. *Journal of Advances in Modeling Earth Systems*, 15(10), e2022MS003444.

PRESENTATIONS

- (Oral) **S. L. Ferretti**, J. Lin, S. Shamekh, ...& T. Beucier (2026). Data-Driven Integration Kernels for Interpretable Nonlocal Operator Learning. *Climate Informatics 2026*. Lausanne, VD, CH.
- (Invited Oral) **S. L. Ferretti**, J. Lin, S. Shamekh, ...& T. Beucier (2025). Making Data-Driven Climate Prediction Interpretable by Design. *Learning the Earth with Artificial Intelligence and Physics Lectures in Climate Data Science*. Virtual.
- (Oral) **S. L. Ferretti**, T. Beucier, M. S. Pritchard, ...& J. W. Baldwin (2025). Learning Nonlocal Controls on South Asian Monsoon Rainfall. *American Geophysical Union Fall Meeting*. New Orleans, LA, USA.
- (Poster) **S. L. Ferretti**, M. S. Pritchard, & J. W. Baldwin (2025). Towards Data-Driven Discovery of Thermodynamic Controls on South Asian Monsoon Rainfall. *Learning the Earth with Artificial Intelligence and Physics Year 5 Meeting*. Briarcliff Manor, NY, USA.
- (Poster) **S. L. Ferretti**, L. Peng, J. W. Baldwin, & M. S. Pritchard (2024). Spatiotemporal Stress-Testing of a Process-Oriented Diagnostic for Rainfall. *American Geophysical Union Fall Meeting*. Washington DC, USA.
- (Poster) **S. L. Ferretti**, J. W. Baldwin, N. Liu, ...& M. S. Pritchard (2024). Unraveling CMIP6 Biases in Summer Monsoon Rainfall Over the Arabian Sea and Western India. *104th American Meteorological Society Annual Meeting*. Baltimore, MD, USA.
- (Poster) **S. L. Ferretti**, N. Liu, J. W. Baldwin, & M. S. Pritchard (2022). Understanding the Monthly Variation in the Upstream Enhancement of Indian Summer Monsoon Precipitation Near the Western Ghats. *American Geophysical Union Fall Meeting*. Chicago, IL, USA.
- (Invited Oral) **S. L. Ferretti** (2022). Using Models to Understand Climate Change. *Making Modern Science*, University of California, Irvine. Irvine, CA, USA.

(Poster) **S. L. Ferretti**, B. Dalton, L. D. Schiferl, ...& R. Commane (2020). Assessing the Meteorological Impact on Methane (CH₄) Emission Changes in New York City During the COVID-19 Shutdown. *American Geophysical Union Fall Meeting*. Virtual.

AWARDS & FELLOWSHIPS

Advancing Inclusive Excellence Award School of Physical Sciences, University of California, Irvine	Jan 2023 - Jun 2023 \$ 400
Diversity, Equity, & Inclusion Graduate Leaders Fellowship School of Physical Sciences, University of California, Irvine	Sep 2022 - Jun 2023 \$ 24,720
Class Award Cornell University Class Council	Apr 2018

TEACHING

University of California, Irvine, Air Quality Management Teaching Assistant	Apr - Jun 2024 Irvine, CA
University of California, Irvine, Earth System Chemistry Teaching Assistant	Sep - Dec 2023 Irvine, CA
University of California, Irvine, Diversity in STEM Seminar Coordinator	Sep - Dec 2022 Irvine, CA

SERVICE & ENGAGEMENT

Climate Informatics Hackathon , Participant	Apr 2026
Journal of Geophysical Research Atmospheres , Reviewer	Nov 2025–Present
UCI ESS First-Year Mentoring Program , Mentor	Sep 2022–Aug 2024
UCI Climate Justice Initiative , Fellowship Application Reviewer	Jun 2023
UCI CLEWS Climate-Tech Seminar Series , Panelist	Jun 2023
UCI ESS Inclusive Excellence Committee , Organizer	Sep 2022–Jun 2023
Community Earth System Model Tutorial , Boulder, CO	Aug 2022
AGU Bridge Program New Student Orientation , Panelist	Jul 2022
Seed Consultant Group , Volunteer Project Consultant	Mar–May 2022
GREEN Program, Microgrid Systems for Rural Development , Kathmandu, Nepal	Jan 2020
Cornell University Dance Team , Cofounder & Vice President	Oct 2017–Dec 2020
Cornell University College Mentors for Kids , Mentor	Aug 2017–May 2018

RELEVANT COURSEWORK _____

GRADUATE

Machine Learning for Environmental Engineering & Science • Atmospheric Dynamics • Geoscience Modeling & Data Analysis • Environmental Fluid Mechanics & Turbulence • Global Physical Climatology • Humans in the Earth System • Planning for Climate Justice • Communication Skills for Environmental Scholars

UNDERGRADUATE

Atmospheric Thermodynamics & Hydrostatics • Climate Dynamics • Microclimatology • Meteorological Observations & Instrumentation • Synoptic Meteorology • Evolution of the Earth System • Interior of the Earth • Oral Communication • Leading People & Effective Communication